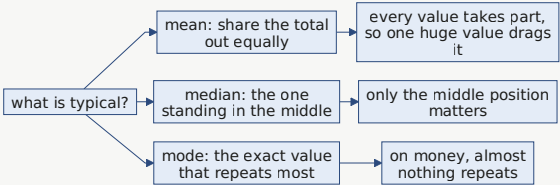


1 the three "typical" numbers



	Answers	Reach for it when
Mean	Total shared out equally	The shape is roughly even
Median	The record in the middle of the queue	Money, durations, anything with a tail
Mode	The value that repeats most	The column is a category

Crux. nobody is right, they answer different questions, and the asker rarely says which one they meant.

2 the one-line skew test

mean / median		
Ratio	Shape	Send
Close to 1	Roughly even	Either
Well above 1	Right tail, something large	Median
Well below 1	Left tail, something small	Median
Run it first on any column you have never seen. It costs one line and it tells you which statistic you are allowed to quote.		

3 what one record does

seven orders	mean Rs 1,676.43	median Rs 1,310
swap the top for 480000	mean Rs 69,842.86	median Rs 1,310
Crux. the mean grew nearly 42 times, the median moved by zero. Not "a little". Zero.		

4 the fence

q1, _, q3 = statistics.quantiles(values, n=4)
iqr = q3 - q1
upper = q3 + 1.5 * iqr
lower = q1 - 1.5 * iqr
Crux. a fence is a flag, never a delete key. It says look at this record. An outlier is a finding to investigate before it is a row to delete.

5 grouping without knowing the categories

```
buckets = {}
for r in orders:
    key = r[field]
    if key not in buckets:
        buckets[key] = {"count": 0, "amounts": [], "returned": 0}
    buckets[key]["count"] += 1
    buckets[key]["amounts"].append(r["amount"])
    if r["status"] == "returned":
        buckets[key]["returned"] += 1
```

Short form of the same idea: counts[key] = counts.get(key, 0) + 1

Crux. every hard-coded list of categories is a promise about data you have not read yet. Hard-code three and Kalpa has four, and you get `KeyError: 'Student'`.

6 count first, then median

Accumulate one record at a time	Needs the whole group first
count, sum, returned count, min, max	median, quartiles, the fence
Crux. collect the amounts into a list during the pass and take the median after the loop ends. There is no running median.	

7 the honest sentence

[the number]	[the denominator]	[what it does not yet support]
Business: returned on 1 of 9 orders, 11.1 percent. Lowest in the file and the smallest segment in it. One more return takes it to 22.2 percent and behind Student.		
Crux. the third part is the one that goes missing between your notebook and somebody else's slide, so it goes in the sentence rather than in a footnote.		

8 before you send a rate

Check	If it fails
Denominator on the same line as the number	It is not ready
Any group under your stated floor	Say so in words, in that line
Ranked groups of very different sizes	Say what the ranking is partly measuring
A mean quoted on a money column	Replace it with the median
A difference being called real	Park it and name what would settle it

THE THREE LINES WORTH MEMORISING

- 1. The mean was right and the description was wrong.
- 2. On a money column, send the median, and say that is what you sent.
- 3. Every rate carries its denominator, or it lies for you while you are not in the room.

Crux. every rate carries its denominator, or it lies for you while you are not in the room.

THE WORDS ON THIS SHEET

Mean The total shared out equally, which every value takes part in	Median The value standing in the middle of the sorted column	Mode The exact value that repeats most, which on money repeats twice in 44	Range Maximum minus minimum, built from the two least typical values	Interquartile range The spread of the middle half, and the basis of the fence
				Skew Which side of the median the tail is on